

Supplementary Material

Interface Band Offsets as the Common Limit Across Four Lead-Free Perovskite Solar-Cell Absorbers: A Coupled Opto-Electrical COMSOL Study

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S1. SUPPLEMENTARY FIGURES

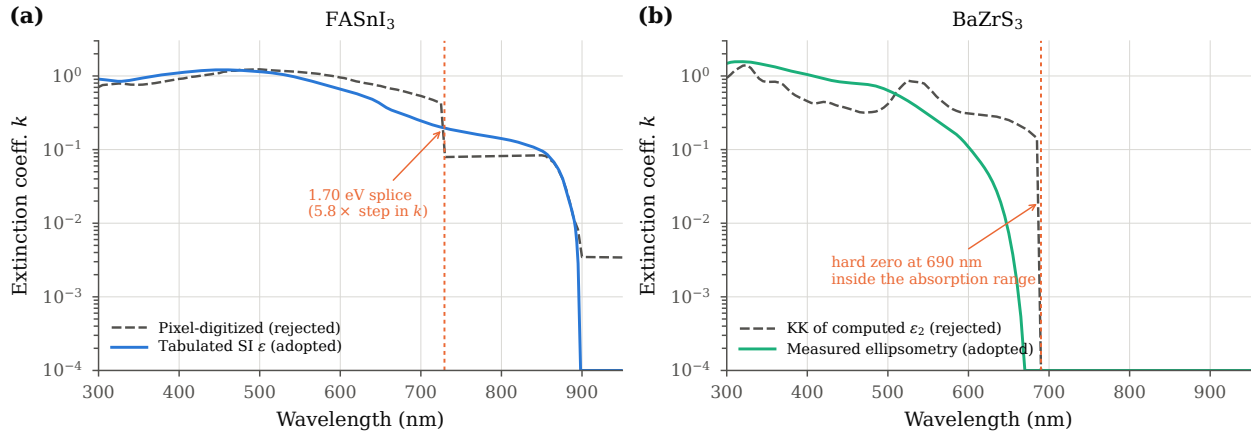


Fig. S1. **Optical-constant source selection.** Extinction coefficient k of (a) FASnI_3 and (b) BaZrS_3 , comparing candidate datasets. (a) A pixel-digitized reconstruction (rejected) splices two traced figures of [1] at 1.70 eV. Across that 5 nm join k falls $0.4263 \rightarrow 0.0792$, a $5.8 \times$ discontinuity, after which it sits nearly flat to 850 nm, behaviour no direct-gap absorber exhibits. The adopted dataset is the tabulated dielectric function from the same paper's supplementary information (695 points, point-by-point inversion of measured ellipsometry). (b) A Kramers-Kronig reconstruction of a *computed* ϵ_2 (rejected) is hard-zeroed at 690 nm, cutting from $\alpha = 2.6 \times 10^4 \text{ cm}^{-1}$ directly to zero inside its own absorption range. The adopted dataset is the measured ellipsometric dielectric function [2].

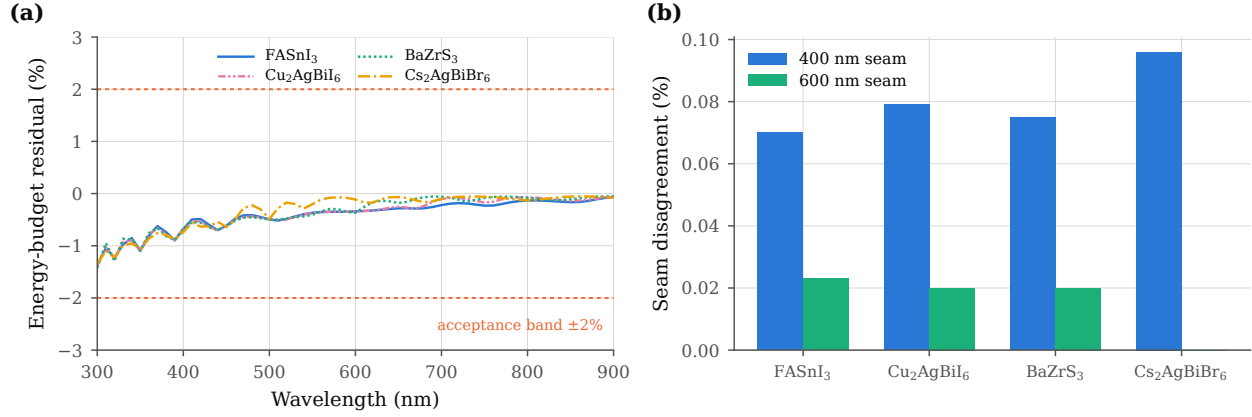


Fig. S2. **Validation.** (a) Per-wavelength energy-budget residual for all four absorbers against the $\pm 2\%$ acceptance band. The worst value across 244 solves is -1.43% , at 300 nm, the finest mesh in the study. (b) Band-seam agreement: wavelengths at the 400 and 600 nm band boundaries solved on both neighbouring meshes, which are generated independently. Agreement is 0.020 to 0.096 %. This is the check that detects the Floquet periodic-mesh defect of Section S4. The energy residual does not.

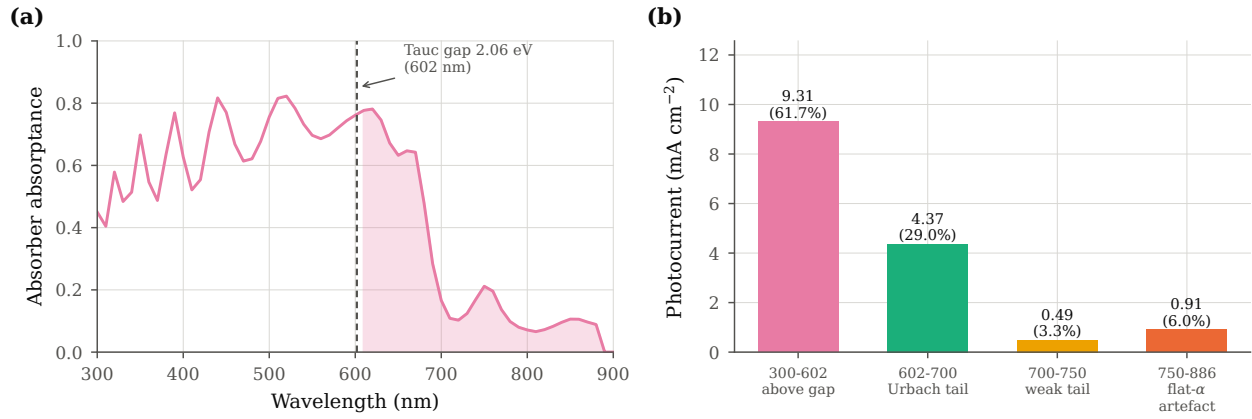


Fig. S3. **Cu₂AgBiI₆ sub-gap absorption.** (a) Absorber absorbance with the 2.06 eV Tauc gap marked. The shaded region is sub-gap. (b) Decomposition of the $15.083 \text{ mA cm}^{-2}$ optical ceiling by spectral region, with the segments partitioning the total exactly. 61.7 % lies above the Tauc gap, 29.0 % is an Urbach-like tail with characteristic energy $\approx 88 \text{ meV}$, 3.3 % is a weak tail, and 6.0 % (0.908 mA cm^{-2}) arises where α stops decaying and sits flat at $\approx 2.7 \times 10^3 \text{ cm}^{-1}$ for 130 nm before stepping to zero at 886 nm, a construction artifact of the optical file rather than a physical tail.

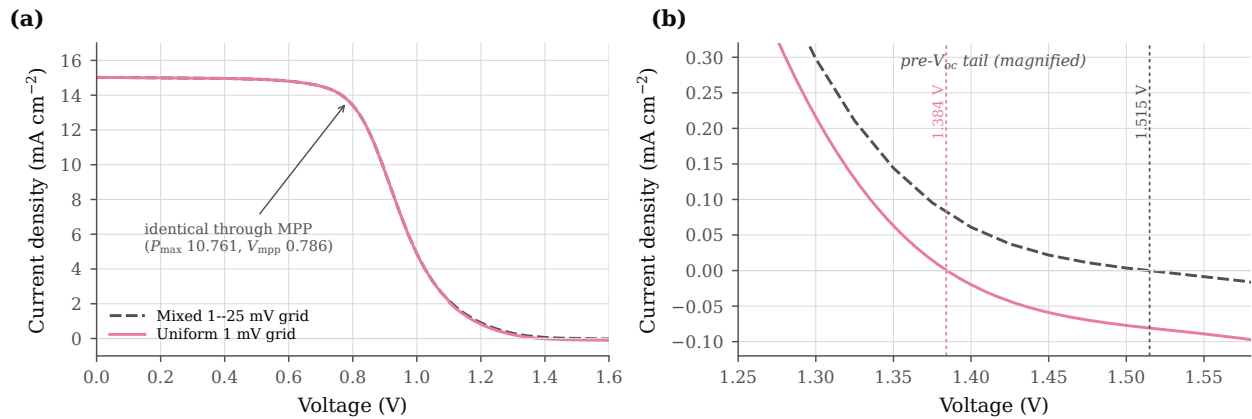


Fig. S4. **Cu₂AgBiI₆ open-circuit voltage is not grid-converged.** (a) J - V on a mixed 1 to 25 mV continuation grid versus a uniform 1 mV grid. The curves are identical through the maximum power point ($P_{\text{max}} = 10.761 \text{ mW cm}^{-2}$, $V_{\text{mpp}} = 0.786 \text{ V}$, $J_{\text{mpp}} = 13.691 \text{ mA cm}^{-2}$, to five significant figures). (b) Magnified pre- V_{oc} tail: the curves diverge only beyond $\approx 1.3 \text{ V}$ where $J < 0.2 \text{ mA cm}^{-2}$ and the characteristic is nearly flat, so a small numerical difference moves the zero crossing from 1.515 V to 1.384 V.

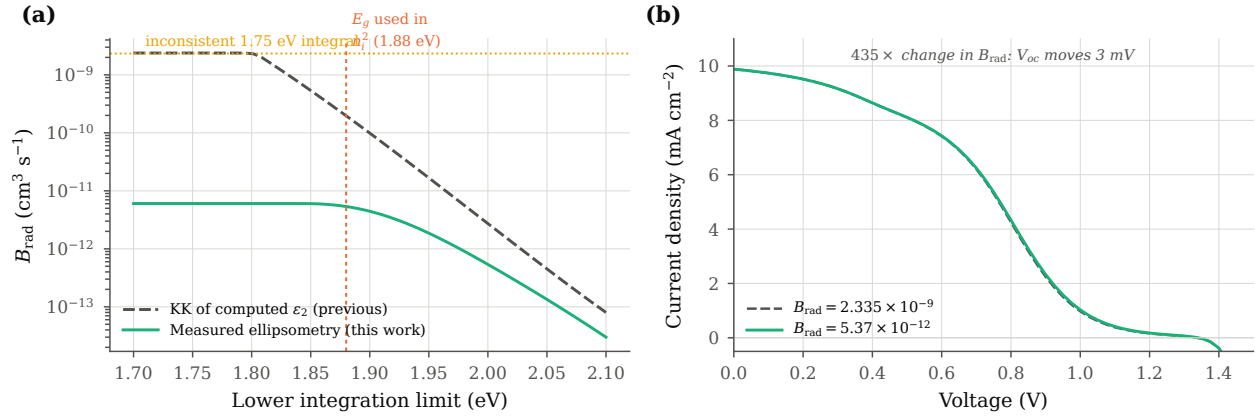


Fig. S5. **BaZrS₃ radiative coefficient.** (a) van Roosbroeck-Shockley B_{rad} versus the lower integration limit, for the rejected (computed) and adopted (measured) optical constants. A candidate value of $2.335 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$ corresponds to the rejected optics integrated from 1.75 eV while n_i^2 is built from $E_g = 1.88 \text{ eV}$, which is internally inconsistent and $\approx 12\times$ too large even on its own data. Evaluated consistently at 1.88 eV on the measured optics, $B_{\text{rad}} = 5.37 \times 10^{-12} \text{ cm}^3 \text{ s}^{-1}$ (adopted). (b) Device J - V with both coefficients: a $435\times$ change moves V_{oc} by 3 mV, establishing that BaZrS₃'s V_{oc} is not radiatively limited.

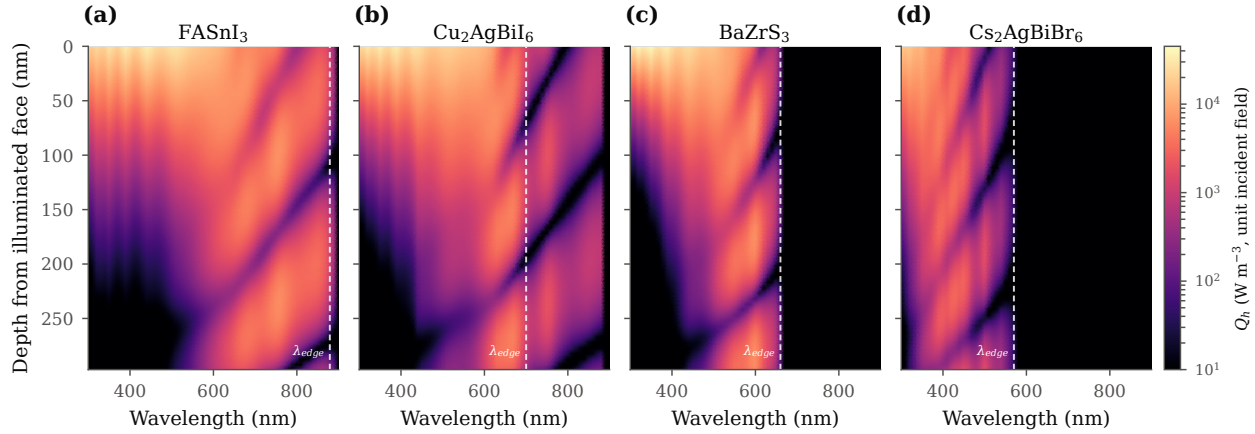


Fig. S6. **Depth- and wavelength-resolved generation.** Absorbed power density $Q_h(z, \lambda)$ through the 300 nm absorber under unit incident field, plotted from the illuminated face. Each absorber's band edge is directly visible as the wavelength beyond which generation ceases: $\approx 880 \text{ nm}$ for FASnI₃, $\approx 700 \text{ nm}$ for Cu₂AgBiI₆ with its sub-gap tail continuing weakly to 900 nm , $\approx 660 \text{ nm}$ for BaZrS₃, and $\approx 570 \text{ nm}$ for Cs₂AgBiBr₆. The diagonal fringes are Fabry-Pérot interference in the finite-thickness stack, which a Beer-Lambert treatment cannot reproduce. The contrast between panels explains the collection behaviour reported in the main text: FASnI₃ generates strongly across the full depth, so an optical improvement that front-loads generation moves carriers into a region it collects less efficiently, whereas Cs₂AgBiBr₆ generates only in a narrow band near the illuminated face regardless.

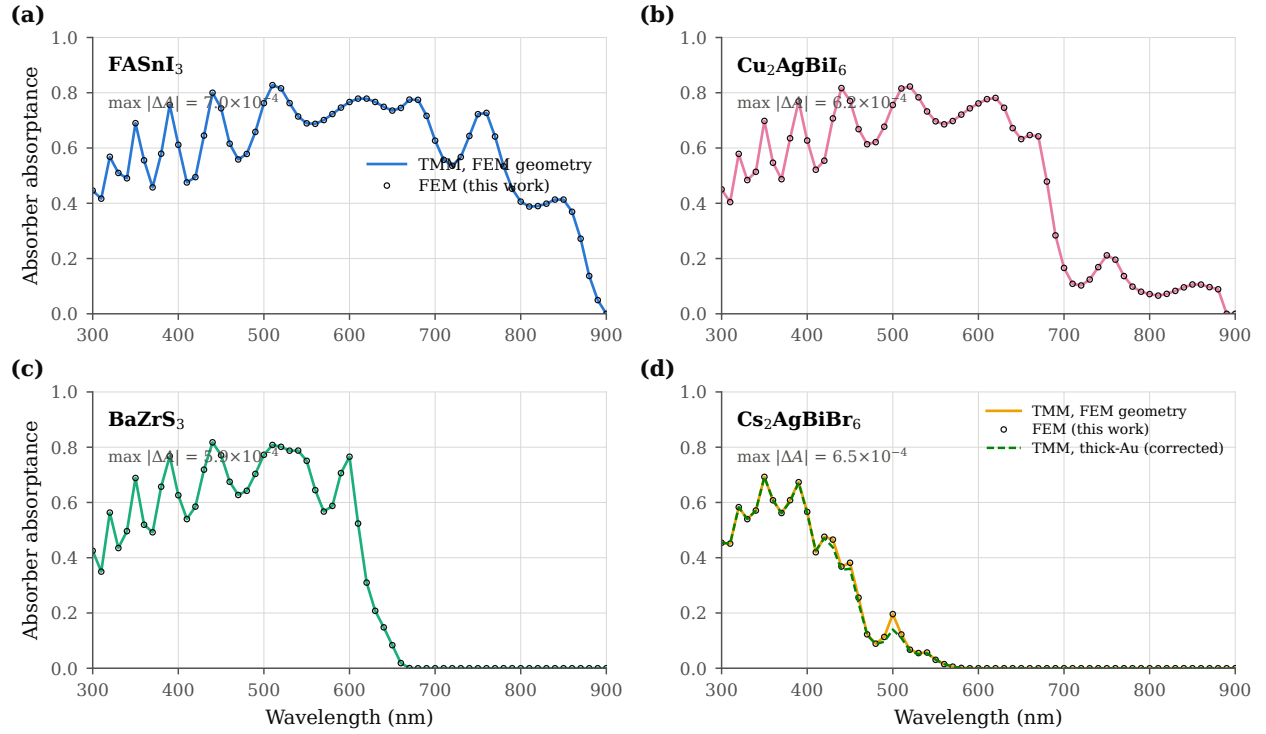


Fig. S7. **Transfer-matrix verification of the electromagnetic solver.** Absorber absorbance from the finite-element solve (open circles) against an independent scalar transfer-matrix calculation with identical optical inputs and identical geometry (solid lines), for all four absorbers. Worst-case disagreement across all 4×61 solves is 7×10^{-4} in absorbance and 0.02 % in AM1.5G photocurrent. For Cs₂AgBiBr₆ the dashed curve shows the thick-contact (semi-infinite Au) limit: the visible departure in the 430 to 570 nm window is the back-contact truncation artifact of Section S3, which the identical-geometry agreement isolates as geometric rather than numerical.

S2. DEVICE ARCHITECTURE AND PARAMETER PROVENANCE

The stack is fixed for all four devices: glass / FTO (600 nm) / TiO₂ (50 nm) / absorber (300 nm) / Spiro-OMeTAD (200 nm) / Au. The optical unit cell is $350 \times 350 \text{ nm}^2$ with Floquet periodicity on all lateral faces.

Sources. Absorber optics: FASnI₃ [1], Cu₂AgBiI₆ [3], [4], BaZrS₃ [2], Cs₂AgBiBr₆ [5]. Gaps and masses: BaZrS₃ [6], band alignment [7], and Cs₂AgBiBr₆ from [5], [8]. Lifetimes: BaZrS₃ 1 ns from single-crystal transient photoluminescence [9] and 33 ns computed from the sulfur-interstitial capture coefficients of [10]. FASnI₃ electron affinity: experimental photoelectron spectroscopy in air (PESA) [11], with the 4.0 eV sweep endpoint from [12]. Transport layers: [13], [14]. Contact optics: FTO [15], TiO₂ [16], Au [17].

Provenance codes. [EXP] experimental. [THEO] computed, level stated. [SIM-LIT] from published device simulations. [DERIVED] computed here from literature inputs by a stated parameter-free relation. [SWEPT] treated as a sensitivity axis.

TABLE S1
ABSORBER PARAMETERS

Quantity	BaZrS ₃	FASnI ₃	Cs ₂ AgBiBr ₆	Cu ₂ AgBiI ₆	Status
E_g (eV)	1.88	1.41	2.32	2.03 ^a	[THEO]/[EXP]
χ (eV)	4.1 (3.9, 4.0)	3.50 (3.75, 4.0)	4.0 (3.79, 4.19)	3.22 (3.16, 3.36)	[SWEPT] ^b
m_e^*/m_0	0.3	0.3	0.33	0.6	[THEO]
m_h^*/m_0	0.9	0.9	0.84	1.0	[THEO]
τ_{SRH}	1/33 ns	43/123 ns	13.7 ns/1 μ s	33/3.3 ns	[SWEPT]
ε_r	79.4	8.2 (6, 10)	5.8 (3.54)	6.3 (12)	[SWEPT] ^b
μ_n, μ_p (cm ² V ⁻¹ s ⁻¹)	37, 11	67 (22)	0.57 (6)	1.7 (5.1)	[SWEPT] ^b
Doping (cm ⁻³)	N_d 4.4×10^{10}	N_a 1.5×10^{17}	N_a 1.42×10^{12}	N_a 1×10^{12}	[SIM-LIT]
B_{rad} (cm ³ s ⁻¹)	5.37×10^{-12}	7.99×10^{-8}	3.19×10^{-12}	1.51×10^{-10}	[DERIVED]

All values read back from the saved models. ^bFormat: base value (alternates swept). The near-intrinsic BaZrS₃ donor density is the deliberate literature value for optimized low-defect processing. The conventional high-temperature-sulfurization range (10^{19} to 10^{20} cm^{-3}) exceeds N_c and is incompatible with the Boltzmann statistics used here.

^aTransport-model value, read back from the saved model. the 2.06 eV Tauc gap of [4] is used for the sub-gap decomposition of Fig. S3.

TABLE S2
TRANSPORT-LAYER AND CONTACT PARAMETERS

Quantity	TiO ₂ (ETL)	Spiro-OMeTAD (HTL)	Source
E_g (eV)	3.2	2.9	[13], [14]
χ (eV)	4.15	2.15	[13], [14]
ε_r	n/a	3	[13]
μ_n, μ_p (cm ² V ⁻¹ s ⁻¹)	n/a	1×10^{-4}	[13]
N_c, N_v (cm ⁻³)	$2.2 \times 10^{18}, 1 \times 10^{19}$	$2.2 \times 10^{18}, 2.2 \times 10^{18}$	[13], [14]
Doping (cm ⁻³)	N_d 5×10^{17}	N_a 1×10^{18}	[13]
Optical	Sellmeier, $k = 0$ [16]	$n = 1.75, k = 0$	Sec. S6
FTO work function (eV)	4.6		[13]

Disclosed disagreements. Two literature conflicts are carried openly rather than silently resolved. Spiro-OMeTAD band gap: 2.9 eV [13] versus 3.2 eV [18]. Spiro N_v : 2.2×10^{18} [13] versus 1.8×10^{19} [18]. We use [13] throughout. A third reported value, a Spiro hole mobility of $2 \times 10^4 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ [18], is rejected as a transcription error: it is roughly ten orders of magnitude above the well-established $\sim 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ for this material. Both disagreements and the rejection were confirmed against each paper's own tables. The HTL acceptor doping is the saved-model value ($1 \times 10^{18} \text{ cm}^{-3}$). See the reproducibility caution below.

A reproducibility caution. The HTL acceptor doping is $1 \times 10^{18} \text{ cm}^{-3}$ in the saved model but $1 \times 10^{19} \text{ cm}^{-3}$ in the build script that nominally creates it. A later stage changed the value and only the model file records it. The tenfold difference eases hole extraction enough to shift BaZrS₃'s fill factor from 0.34 to 0.38, so parameters must be read back from the saved model, never from the build script.

A. Gap consistency across the three roles a gap plays

A band gap is used in three distinct roles here (the transport model's n_i^2 , the detailed-balance ceiling, and the absorption onset of the optical constants) and these are not automatically the same number. Measured directly from the computed spectra:

TABLE S3
BAND GAP USED IN EACH ROLE, AND THE RESULTING SPREAD

Absorber	transport n_i^2	DB ceiling	optical onset	spread
FASnI ₃	1.41	1.41	1.39	0.02 eV
BaZrS ₃	1.88	1.94	1.88	0.06 eV
Cs ₂ AgBiBr ₆	2.32	2.10	2.21	0.22 eV
Cu ₂ AgBiI ₆	2.03	2.06	1.41	0.65 eV

FASnI₃ and BaZrS₃ are consistent. **Cs₂AgBiBr₆ is not:** its HSE06 transport gap (2.32 eV), the 2.10 eV used for its detailed-balance ceiling, and the 2.21 eV onset of its own optical constants disagree by up to 0.22 eV. The device consequence was tested directly by re-solving the device with $E_g = 2.21$ eV (its own optical onset): J_{sc} , P_{max} and PCE are unchanged to four decimal places (1.170 %), while the gap shift appears entirely in V_{oc} (-111 mV $\approx \Delta E_g$, as n_i^2 dictates) with the fill factor compensating exactly ($0.281 \rightarrow 0.300$) because the maximum power point sits on the barrier-limited plateau at 0.60 V. The same test on BaZrS₃ ($1.88 \rightarrow 1.94$ eV) gives the same structure: PCE unchanged at 4.499 %, $V_{oc} + 62$ mV, FF $0.340 \rightarrow 0.325$. Efficiency-level conclusions are therefore invariant to the gap choice. Only V_{oc} carries it, by a predictable $\approx \Delta E_g$. **Cu₂AgBiI₆'s 0.65 eV spread is by construction**, since its optical file carries a deliberate sub-gap tail to 1.40 eV. The consequence there is material and is discussed in the main text.

S3. THE BACK-CONTACT TRUNCATION ARTIFACT IN CS₂AGBIBR₆

The optical model truncates the Au contact at 45 nm backed by a perfect electric conductor, a deliberate cost measure justified by Au's 12 to 17 nm near-infrared skin depth (three-plus skin depths of metal, below 10^{-6} returning power). The justification fails in one window: Au's interband damping dips near 450 to 570 nm, roughly doubling the skin depth, and Cs₂AgBiBr₆ is the one absorber nearly transparent exactly there. Light reaches the back contact, and the PEC backing returns an unphysically strong wave: at 510 nm the truncated geometry gives $R = 0.469$ while realistically thick Au gives $R = 0.292$. The other three absorbers extinguish this window before it reaches the contact and agree with the thick-contact limit to ≤ 0.35 % in photocurrent.

The correction uses the transfer-matrix method of Fig. S7, which is exact for this planar stack and validated against the finite-element solver to 7×10^{-4} on identical geometry. Three gates were passed in order: (i) the PEC-geometry TMM reproduces the FEM spectrum to 9×10^{-4} across absorptance and reflectance jointly (6.5×10^{-4} on absorber absorptance alone, the metric of Table S5 and Fig. S7). (ii) The same pipeline reproduces the FEM generation profile $G(z)$, including the AM1.5G bin weighting, to 0.10 % with identical photocurrent. (iii) Only then was the thick-contact spectrum and $G(z)$ emitted and the Cs₂AgBiBr₆ transport re-solved on it. Corrected numbers: ceiling $2.546 \rightarrow 2.440$ mA cm⁻² (the truncated value overstated the ceiling by 4.3 %), device J_{sc} $2.473 \rightarrow 2.371$ mA cm⁻², PCE $1.22 \rightarrow 1.17$ %. Collection is unchanged (97.2 %). The 45 nm-truncation figure of 2.546 appears nowhere in the main text except as the disclosed pre-correction value.

S4. THE FLOQUET PERIODIC-MESH DEFECT

Floquet periodic boundary conditions require the paired faces of the unit cell to carry **identical surface meshes**. A free-tetrahedral mesher does not produce this. The solver does not raise an error. It interpolates across the mismatch, and the computed per-layer absorption depends on the particular mesh realization. Measured on this stack at 700 nm:

A 36 % swing for a 2.4 % change in element count. The remedy is to mesh $x = 0$ with a free-triangular operation, copy that mesh onto $x = L$, repeat for y , and only then fill with tetrahedra. Feature order matters because the x and y faces share the cell's vertical edges. After the fix the same comparison gives 0.783103 versus 0.782888 (0.03 %). The global energy budget also improves, from 1.2 % to 0.03 %. Note, however, that even the defective mesh sat comfortably inside the ± 2 % acceptance band while the layer partition was wrong by 36 %: the residual shifts, but far too weakly to be diagnostic, which is precisely why it cannot serve as the detector.

TABLE S4
ABSORBER ABSORPTANCE VERSUS MESH REALIZATION, BEFORE THE FIX

Elements	A_{absorber}
120 153	0.719
120 346	0.663
123 031	0.463

The energy-conservation residual does not detect this defect. It compares total absorbed power against total incident flux and is insensitive to how that total is partitioned between layers, which is precisely what the photocurrent depends on. Detection requires either a mesh-versus-mesh comparison or a tangential-field equality check on the paired faces (which reaches 0.00000 % after the fix). Two traps arise in the latter. Compare the *tangential* field components, not $|E|$, because the normal component sampled on a face is interpolated from whichever side the evaluator picks, and expect the x and y pairs to differ, because the incident wave is x -polarized.

S5. NUMERICAL PROCEDURE

Wavelength banding. Element count scales as λ^{-3} , so meshes are generated per band: 300 to 390, 400 to 590, 600 to 900 nm. Element size is $\lambda/(n_{\text{eff}} \text{ ppw})$ with $n_{\text{eff}} = |\tilde{n}|$, subject to a skin-depth clamp and a floor of three elements across any layer thickness. The wavelength-independent cap is applied only to air and the perfectly matched layer, never to the absorber.

Two independent photocurrent routes. The depth integral of $G(z)$ and the spectral integral of layer absorptance are computed separately and compared every run. They share the solve but nothing else. Agreement across the four absorbers is +0.05, +0.03, +0.04, −0.04 %.

AM1.5G quadrature. Irradiance is integrated over each wavelength bin rather than sampled at bin centres, and the outer bin edges are clamped to the grid endpoints. Bin-centre sampling biased the result by $\approx 10\%$ and unclamped edges by a further +23 % on a coarse grid. Both are quadrature errors rather than physics, and both were exposed by the two-route comparison.

Bias continuation. For $\text{Cu}_2\text{AgBiI}_6$ a 25 mV grid converges onto spurious S-shaped negative-differential-resistance branches reporting $\text{FF} = 4.13$ and $\text{PCE} = 86.6\%$. A 1 mV grid removes them. Only dense results are quoted for that absorber. The artifact is reproducible rather than stochastic.

TABLE S5
VALIDATION LEDGER

Check	FASnI_3	$\text{Cu}_2\text{AgBiI}_6$	BaZrS_3	$\text{Cs}_2\text{AgBiBr}_6$
Optical budget closure	−0.00 %	+0.00 %	+0.00 %	+0.00 %
Worst energy residual	−1.36 %	−1.39 %	−1.43 %	−1.35 %
$G(z)$ vs spectrum	+0.05 %	+0.03 %	+0.04 %	−0.04 %
Band seam, 400 nm	0.070 %	0.079 %	0.075 %	0.096 %
Band seam, 600 nm	0.023 %	0.020 %	0.020 %	0.000 %
TMM vs FEM, max $ \Delta A $ (identical geometry)	7.0×10^{-4}	6.2×10^{-4}	5.9×10^{-4}	6.5×10^{-4}

S6. OPTICAL MODEL LIMITATIONS

Lossless transport layers. TiO_2 is modeled with a Sellmeier real index (measured on rutile, while ETL films are typically anatase, a further approximation of this treatment) clamped below 400 nm (its 283 nm pole otherwise returns $n = 5.58$ at the blue end of the grid), and Spiro-OMeTAD with a constant $n = 1.75$. Both carry $k = 0$. Their budget entries are therefore exactly zero and all front parasitic absorption is attributed to the FTO. TiO_2 genuinely absorbs below its ≈ 390 nm edge, so UV generation in the absorber is slightly overestimated. The treatment is identical for all four devices, so relative comparisons are unaffected.

FTO. Previously modeled with a fabricated flat $k = 0.02$. Here it uses measured $\text{SnO}_2\text{:F}$ dispersion.

Gold contact. Truncated to 45 nm in the optical model, more than three skin depths at all wavelengths considered and backed by a perfect electric conductor, while the transport model retains the full 80 nm. The generation profile is therefore shifted by 35 nm when handed to the transport stage, and both frames are provided in the archive.

S7. THE $\text{Cu}_2\text{AgBiI}_6$ SUB-GAP COLLECTION BRACKET

38.2% of $\text{Cu}_2\text{AgBiI}_6$'s photocurrent originates below its Tauc gap (Fig. S3), and the transport model collects it like band-to-band generation. To bound what that assumption is worth, the device was re-solved on two restricted generation profiles built from the per-wavelength absorbed-power data (the full reconstruction reproduces the production profile exactly before any restriction is applied): one with only above-gap bins ($\lambda \leq 600$ nm), and one with all bins except the 750 to 880 nm segment where the absorption coefficient sits at an unphysical flat floor, a construction artifact of the optical file rather than a real Urbach tail (these are the 10 nm solve bins covering the flat segment, whose true edge in the optical file lies at ≈ 885 nm. The removed bins carry 0.99 mA cm^{-2} versus the 0.91 mA cm^{-2} continuum integral of Fig. S3, the difference being bin quantization). The same quantization applies to the above-gap member: its binned profile ($\lambda \leq 600$ nm bins) integrates to 9.484 mA cm^{-2} versus the 9.31 mA cm^{-2} continuum integral of Fig. S3, and the device's 9.442 mA cm^{-2} is 99.6% of its own binned profile, not of the continuum figure, so no entry exceeds its ceiling.

TABLE S6
 $\text{Cu}_2\text{AgBiI}_6$ SUB-GAP COLLECTION BRACKET (UNIFORM 1 MV GRID)

Generation	J_{sc}	V_{oc} (V)	FF	PCE
above-gap only (floor)	9.442	1.367	0.541	6.98 %
artifact-free (defensible)	14.032	1.382	0.521	10.10 %
full collection (as Table I, main text)	15.017	1.384	0.518	10.76 %

The decomposition separates two very different quantities: the construction artifact alone is worth 0.66 absolute PCE points, and the assumption that genuine tail states collect like band carriers a further 3.12 points. The artifact-free 10.10% is the defensible central value. Table I of the main text retains the full-collection number for cross-absorber consistency, with this bracket qualifying it.

S8. INTERFACE RECOMBINATION IS NOT REPRESENTABLE IN THIS FORMULATION

No experimental interface-recombination velocity exists for any of these absorber/transport-layer pairs, and the efficiencies in the main text correspond to defect-free interfaces and are therefore upper bounds.

An attempt was made to sweep an interface recombination velocity $S = 10^2$ to 10^6 cm s^{-1} at the two junctions, and it is reported here as unrepresentable rather than as a result: the solver's trap-assisted surface-recombination boundary feature, offered every boundary in the geometry, accepts only exterior and lateral boundaries and excludes the two interior heterojunction boundaries entirely, silently solving with an empty applicable selection if assigned to them. Two consequences are drawn. (i) All efficiencies here are explicitly labelled defect-free upper bounds, with no interface-sensitivity claim attached. (ii) Because the limiting mechanism in every device is carrier accumulation at a band-offset cliff, real interface recombination would act on precisely the populations this model shows piling up at those interfaces. The upper-bound label is therefore load-bearing, not a formality. Representing junction recombination faithfully would require a different construction (for example, thin defective interlayers), which we leave to future work.

S9. DATA AND CODE AVAILABILITY

All inputs, scripts, results and figure code are archived at

<https://github.com/mohiuddinsciencelover-glitch/leadfree-pv-3d-simulation> REPRODUCE.md documents the pipeline end to end, including the five traps that cost significant time during development: the periodic-mesh defect, the build-script/model parameter divergence, the $\text{Cu}_2\text{AgBiI}_6$ continuation-grid artifact, the two AM1.5G quadrature errors, and the Au-truncation artifact of Section S3.

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